P2 Chapter 10: Numerical Methods

Iteration

Using iteration to approximate a root

Edexcel C3 Jan 2013

$$g(x) = e^{x-1} + x - 6$$

(*a*) Show that the equation $g(x) = 0$ can be written as

$$x = \ln(6 - x) + 1, \qquad x < 6.$$

(2)

The root of $g(x) = 0$ is α.

The iterative formula

$$x_{n+1} = \ln(6 - x_n) + 1, \qquad x_0 = 2.$$

is used to find an approximate value for α.

(*b*) Calculate the values of x_1, x_2 and x_3 to 4 decimal places.

(3)

(*c*) By choosing a suitable interval, show that $\alpha = 2.307$ correct to 3 decimal places.

(3)

a

?

Tip: The difficulty is that there's multiple choices of x to isolate on one side of the equation. Therefore use the target equation to give clues for how to rearrange.

✎ To solve $f(x) = 0$ by an iterative method, rearrange into a form $x = g(x)$ and use the iterative formula $x_{n+1} = g(x_n)$

We'll see why it works later.

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$$0 = e^{x-1} + x - 6$$
$$6 - x = e^{x-1}$$
$$\ln(6 - x) = x - 1$$
$$\ln(6 - x) + 1 = x$$

Tip: The difficulty is that there's multiple choices of x to isolate on one side of the equation. Therefore use the target equation to give clues for how to rearrange.

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x_0, x_1, x_2 represent successively better approximations of the root, where x_0 is the starting value.

Tip: Initially type x_0 (i.e. 2) onto your calculator.
Now just type:

$$\ln(6 - ANS) + 1$$

And then spam your = key to get successive iterations.

b

?

Exam Tip: Show the substitution for x_1 to ensure you get the method mark. But then just write the final value for x_2 and thereafter, as the remaining marks will be 'accuracy' ones.

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b

$$x_0 = 2$$
$$x_1 = \ln(6 - 2) + 1 = 2.386 \ldots$$
$$x_2 = 2.2847 \ldots$$
$$x_3 = 2.3125$$

Exam Tip: Show the substitution for x_1 to ensure you get the method mark. But then just write the final value for x_2 and thereafter, as the remaining marks will be 'accuracy' ones.

Does the starting value x_0 matter?

Yes! We'll see why in a bit when we look at staircase and cobweb diagrams.

- If there are a multiple roots, iteration might converge to (i.e. approach) a different root.
- Or we may not converge to a root at all, and **diverge** (i.e. approach infinity).

[Textbook] $f(x) = x^3 - 3x^2 - 2x + 5$

(a) Show that the equation $f(x) = 0$ has a root in the interval $3 < x < 4$.

(b) Use the iterative formula $x_{n+1} = \sqrt{\frac{x_n^3 - 2x_n + 5}{3}}$ to calculate the values of x_1, x_2 and x_3, giving your answers to 4 decimal places, and taking:

(i) $x_0 = 1.5$ (ii) $x_0 = 4$

(a) ?

(b) (ii) ?

(b) i) ?

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(i) $x_0 = 1.5$ (ii) $x_0 = 4$

(a) $f(3) = -1 \quad f(4) = 13$
There is a change in sign in the interval $3 < x < 4$ and f is continuous, so there is a root of $f(x)$ in this interval.

(b) i) $x_1 = \sqrt{\frac{1.5^3 - 2(1.5) + 5}{3}} = 1.3385 \ldots$
$x_2 = 1.2544 \ldots \quad , x_3 = 1.2200 \ldots$
This appears to be convergent as the change in the root on each iteration is becoming smaller.

(b) (ii) $x_1 = 4.5092 \ldots$
$x_2 = 5.4058$
$x_3 = 7.1219 \ldots$

The series is **divergent**, so iteration has failed.

Test Your Understanding

Edexcel C3 June 2012 Q2

$$f(x) = x^3 + 3x^2 + 4x - 12$$

(*a*) Show that the equation $f(x) = 0$ can be written as

$$x = \sqrt{\left(\frac{4(3-x)}{(3+x)}\right)}, \quad x \neq -3. \qquad \textbf{(3)}$$

The equation $x^3 + 3x^2 + 4x - 12 = 0$ has a single root which is between 1 and 2.

(*b*) Use the iteration formula

$$x_{n+1} = \sqrt{\left(\frac{4(3-x_n)}{(3+x_n)}\right)}, \quad n \geq 0,$$

with $x_0 = 1$ to find, to 2 decimal places, the value of x_1, x_2 and x_3. **(3)**

The root of $f(x) = 0$ is α.

(*c*) By choosing a suitable interval, prove that $\alpha = 1.272$ to 3 decimal places. **(3)**

Warning: Any particular mark scheme gives what is <u>minimally</u> acceptable. So you should use the full wording earlier to avoid the risk of mark loss.

(a) ?

(b) ?

(c) ?

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(a) $x^3 + 3x^2 + 4x - 12 = 0 \Rightarrow x^3 + 3x^2 = 12 - 4x$ — M1

$\Rightarrow x^2(x+3) = 12 - 4x$

$\Rightarrow x^2 = \frac{12-4x}{(x+3)} \Rightarrow x = \sqrt{\frac{4(3-x)}{(x+3)}}$ — dM1A1*

(3)

(b) $x_1 = 1.41$, *awrt* $x_2 = 1.20$ $x_3 = 1.31$ — M1A1,A1

(3)

(c) Choosing (1.2715,1.2725) or tighter containing root 1.271998323 — M1

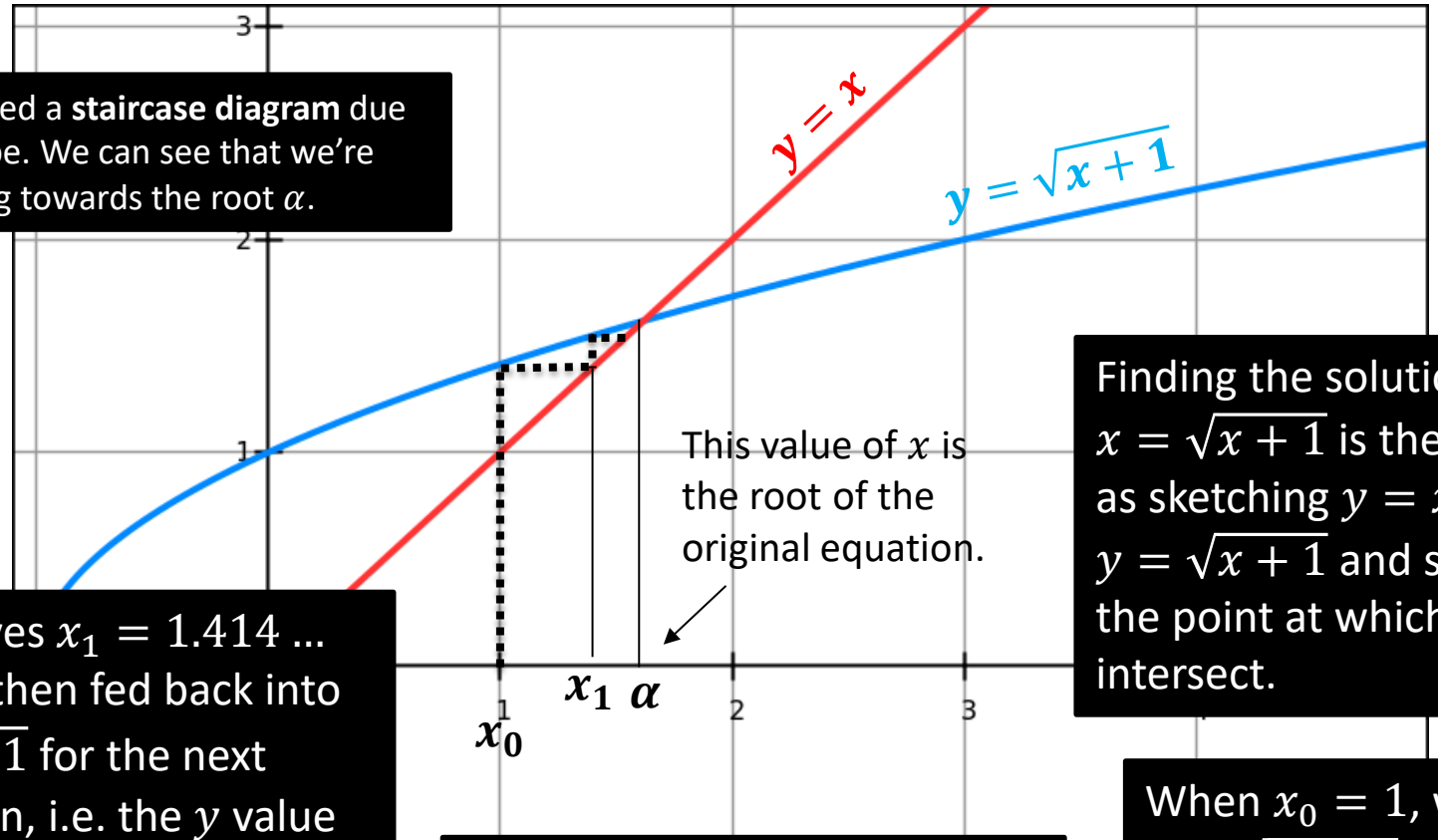
$f(1.2725) = (+)0.00827...$ $f(1.2715) = -0.00821....$ — M1

Change of sign$\Rightarrow \alpha = 1.272$ — A1

Why does this method work?

Solve $x^2 - x - 1 = 0$

Recall we put in the form $x = g(x)$: in this case $x = \sqrt{x+1}$ is one possible rearrangement.
We can then use the recurrence $x_{n+1} = \sqrt{x_n + 1}$. Why does this recurrence work?



This is called a **staircase diagram** due to its shape. We can see that we're converging towards the root α.

This value of x is the root of the original equation.

Finding the solution to $x = \sqrt{x+1}$ is the same as sketching $y = x$ and $y = \sqrt{x+1}$ and seeing the point at which they intersect.

This gives $x_1 = 1.414\ldots$ This is then fed back into $\sqrt{x_n + 1}$ for the next iteration, i.e. the y value becomes the new x value! This is equivalent to moving to the line $y = x$.

We can repeat this process using $x_1 = 1.414\ldots$ to get x_2 and so on.

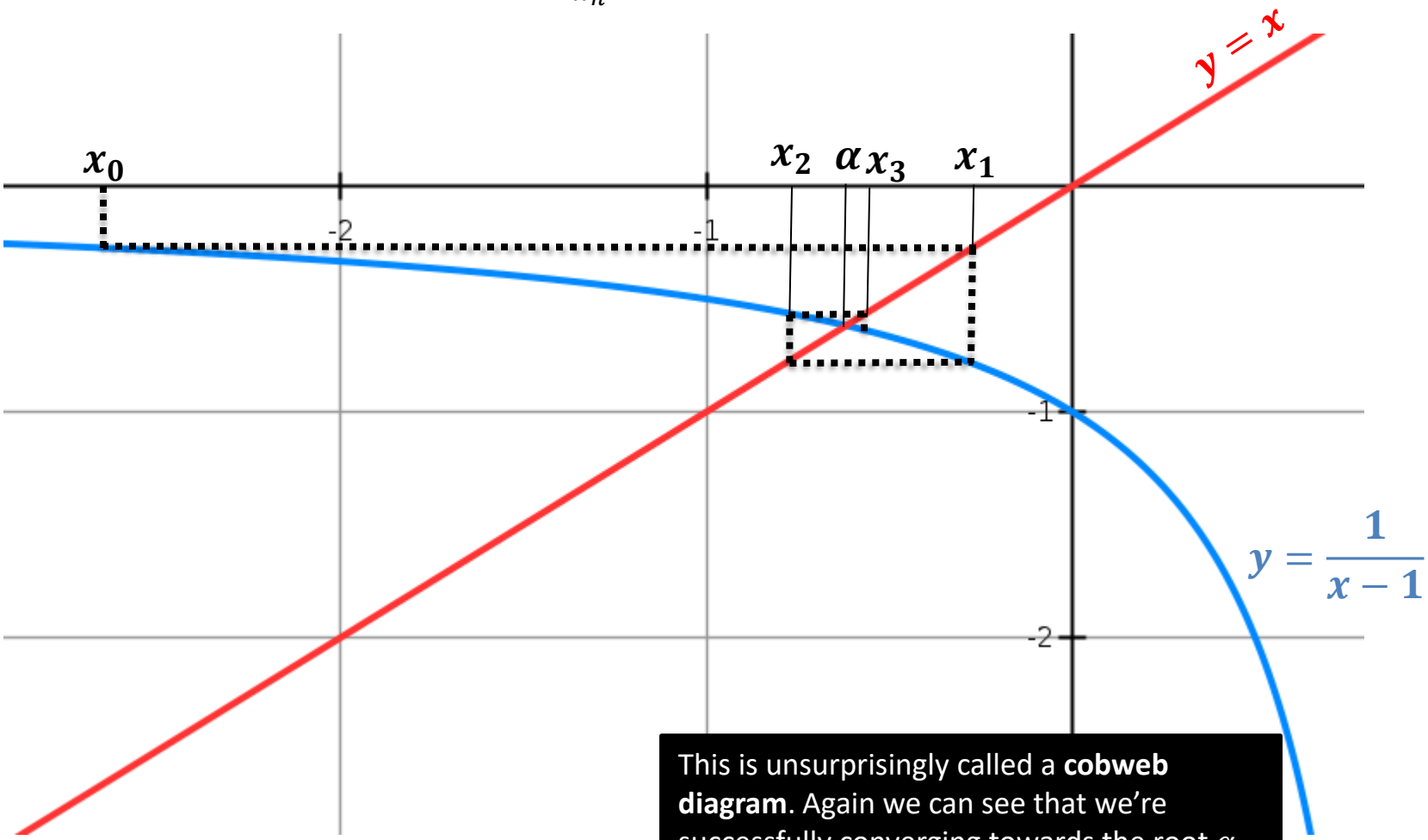
When $x_0 = 1$, we would find $\sqrt{x_0 + 1}$. This is the y value on the $y = \sqrt{x+1}$ graph.

Cobweb Diagrams

Solve $x^2 - x - 1 = 0$

We could also have rearranged differently to $x = \frac{1}{x-1}$

Therefore we use the recurrence $x_{n+1} = \frac{1}{x_n - 1}$. What happens this time?

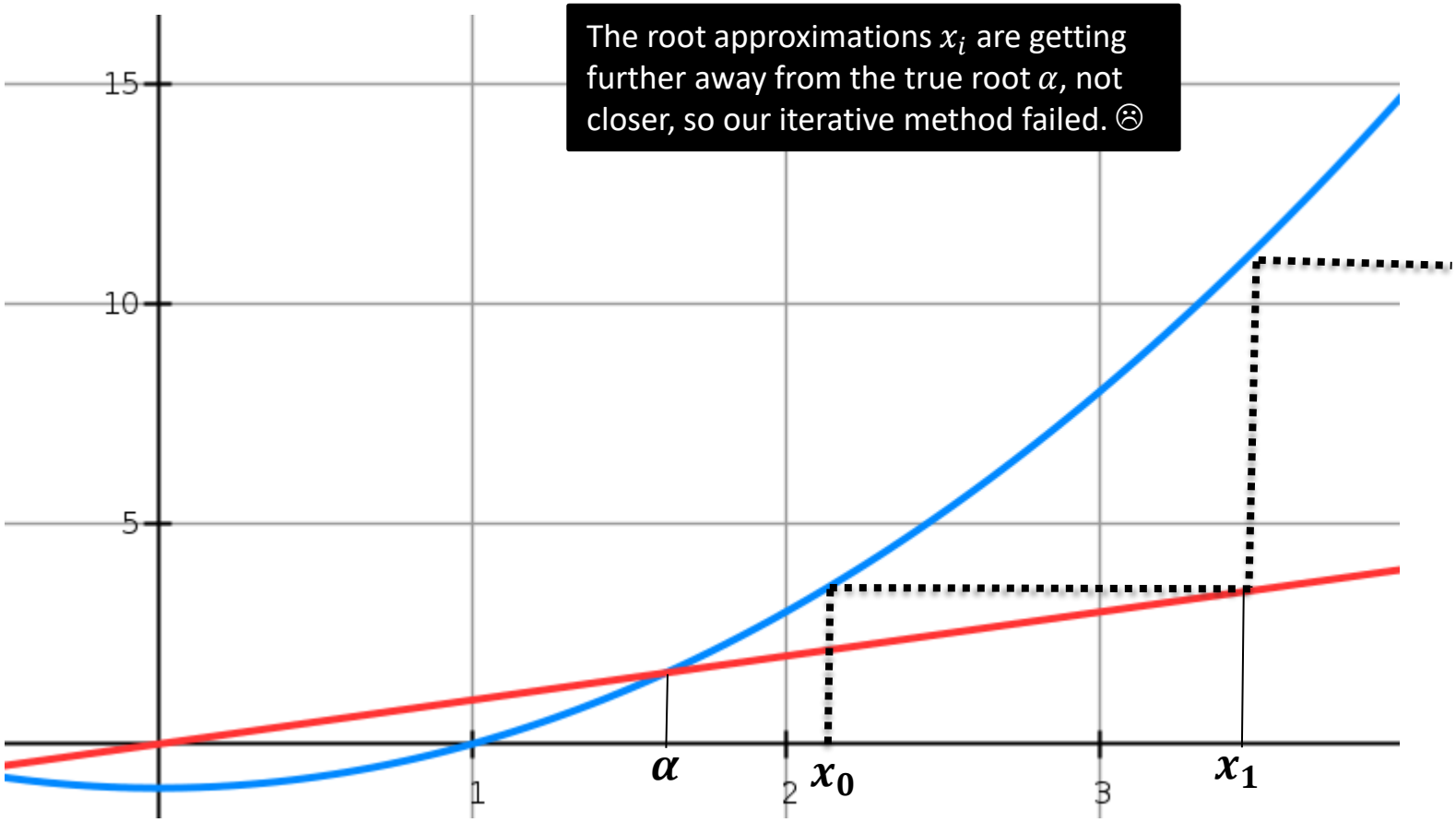


This is unsurprisingly called a **cobweb diagram**. Again we can see that we're successfully converging towards the root α.

And when iteration fails…

Solve $x^2 - x - 1 = 0$

But again, we could have rearranged differently! $x = x^2 - 1$
Using the recurrence $x_{n+1} = x_n^2 - 1$: what happens this time?



Test Your Understanding

$$f(x) = x^2 - 8x + 4$$

(a) Show that the root of the equation $f(x) = 0$ can be written as $x = \sqrt{8x - 4}$

(b) Using the iterative formula $x_{n+1} = \sqrt{8x_n - 4}$, and starting with $x_0 = 1$, draw a staircase diagram, indicating x_0, x_1, x_2 on your x-axis, as well as the root α.

?

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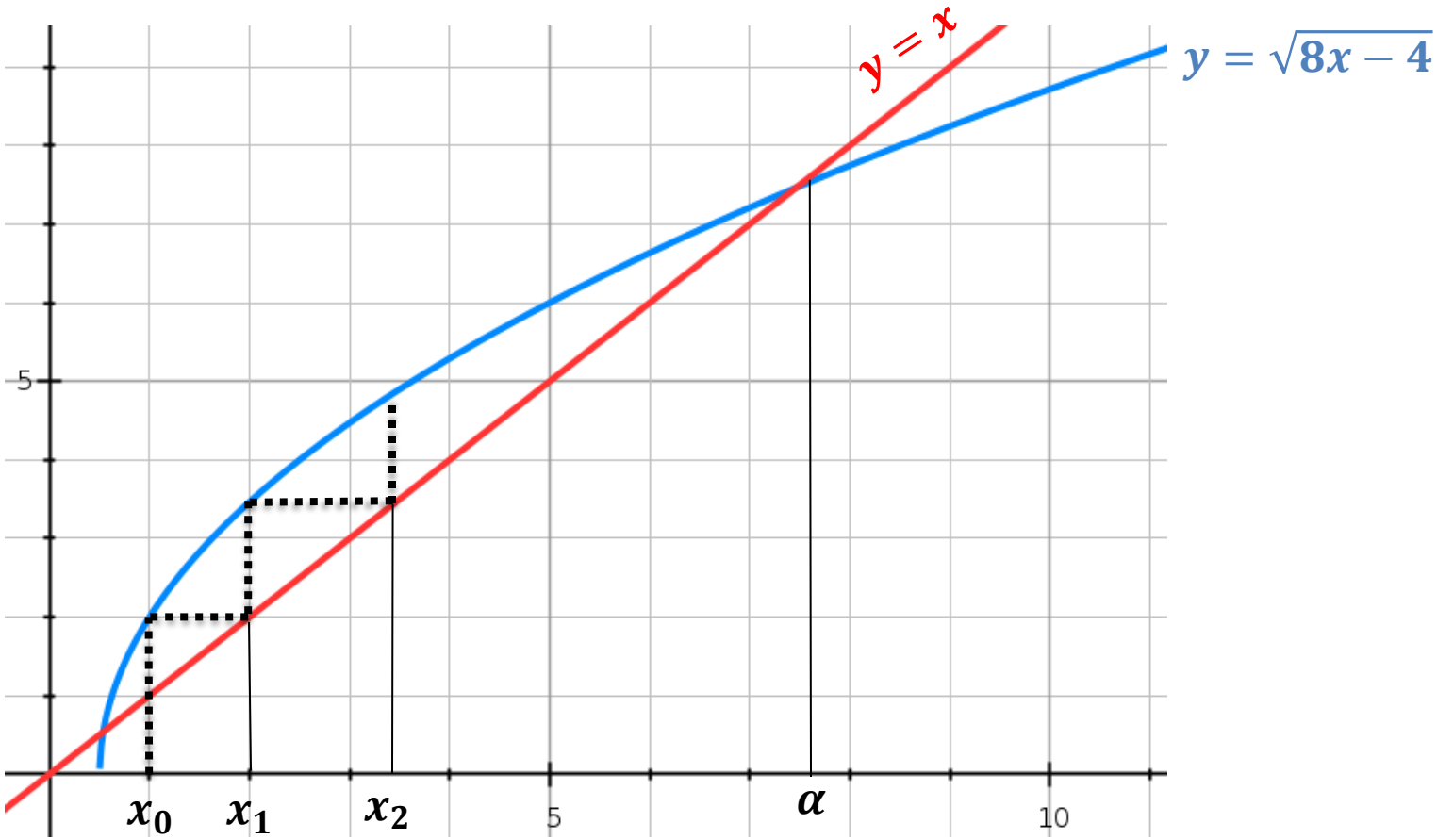

Exercise 10.2

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Homework Exercise

1 $f(x) = x^2 - 6x + 2$

a Show that $f(x) = 0$ can be written as:

i $x = \frac{x^2 + 2}{6}$ **ii** $x = \sqrt{6x - 2}$ **iii** $x = 6 - \frac{2}{x}$

b Starting with $x_0 = 4$, use each iterative formula to find a root of the equation $f(x) = 0$. Round your answers to 3 decimal places.

c Use the quadratic formula to find the roots to the equation $f(x) = 0$, leaving your answer in the form $a \pm \sqrt{b}$, where a and b are constants to be found.

2 $f(x) = x^2 - 5x - 3$

a Show that $f(x) = 0$ can be written as:

i $x = \sqrt{5x + 3}$ **ii** $x = \frac{x^2 - 3}{5}$

b Let $x_0 = 5$. Show that each of the following iterative formulae gives different roots of $f(x) = 0$.

i $x_{n+1} = \sqrt{5x_n + 3}$ **ii** $x_{n+1} = \frac{x_n^2 - 3}{5}$

3 $f(x) = x^2 - 6x + 1$

a Show that the equation $f(x) = 0$ can be written as $x = \sqrt{6x - 1}$. **(1 mark)**

b Sketch on the same axes the graphs of $y = x$ and $y = \sqrt{6x - 1}$. **(2 marks)**

c Write down the number of roots of $f(x)$. **(1 mark)**

d Use your diagram to explain why the iterative formula $x_{n+1} = \sqrt{6x_n - 1}$ converges to a root of $f(x)$ when $x_0 = 2$. **(1 mark)**

$f(x) = 0$ can also be rearranged to form the iterative formula $x_{n+1} = \frac{x_n^2 + 1}{6}$

e By sketching a diagram, explain why the iteration diverges when $x_0 = 10$. **(2 marks)**

Homework Exercise

4 $f(x) = xe^{-x} - x + 2$

a Show that the equation $f(x) = 0$ can be written as $x = \ln\left|\frac{x}{x-2}\right|$, $x \neq 2$.

$f(x)$ has a root, α, in the interval $-2 < x < -1$.

b Use the iterative formula $x_{n+1} = \ln\left|\frac{x_n}{x_n - 2}\right|$, $x \neq 2$ with $x_0 = -1$ to find, to 2 decimal places, the values of x_1, x_2 and x_3.

5 $f(x) = x^3 + 5x^2 - 2$

a Show that $f(x) = 0$ can be written as:

i $x = \sqrt[3]{2 - 5x^2}$ **ii** $x = \frac{2}{x^2} - 5$ **iii** $x = \sqrt{\frac{2 - x^3}{5}}$

b Starting with $x_0 = 10$, use the iterative formula in part **a (ii)** to find a root of the equation $f(x) = 0$. Round your answer to 3 decimal places.

c Starting with $x_0 = 1$, use the iterative formula in part **a (iii)** to find a different root of the equation $f(x) = 0$. Round your answer to 3 decimal places.

d Explain why this iterative formulae cannot be used when $x_0 = 2$.

6 $f(x) = x^4 - 3x^3 - 6$

a Show that the equation $f(x) = 0$ can be written as $x = \sqrt[3]{px^4 + q}$, where p and q are constants to be found. **(2 marks)**

b Let $x_0 = 0$. Use the iterative formula $x_{n+1} = \sqrt[3]{px_n^4 + q}$, together with your values of p and q from part **a**, to find, to 3 decimal places, the values of x_1, x_2 and x_3. **(3 marks)**

The root of $f(x) = 0$ is α.

c By choosing a suitable interval, prove that $\alpha = -1.132$ to 3 decimal places. **(3 marks)**

Homework Exercise

7 $f(x) = 3\cos(x^2) + x - 2$

a Show that the equation $f(x) = 0$ can be written as $x = \left(\arccos\left(\frac{2-x}{3}\right)\right)^{\frac{1}{2}}$ **(2 marks)**

b Use the iterative formula $x_{n+1} = \left(\arccos\left(\frac{2-x_n}{3}\right)\right)^{\frac{1}{2}}$, $x_0 = 1$ to find, to 3 decimal places, the values of x_1, x_2 and x_3. **(3 marks)**

c Given that $f(x) = 0$ has only one root, α, show that $\alpha = 1.1298$ correct to 4 decimal places. **(3 marks)**

8 $f(x) = 4\cot x - 8x + 3$, $0 < x < \pi$, where x is in radians.

a Show that there is a root α of $f(x) = 0$ in the interval $[0.8, 0.9]$. **(2 marks)**

b Show that the equation $f(x) = 0$ can be written in the form $x = \frac{\cos x}{2\sin x} + \frac{3}{8}$ **(3 marks)**

c Use the iterative formula $x_{n+1} = \frac{\cos x_n}{2\sin x_n} + \frac{3}{8}$, $x_0 = 0.85$ to calculate the values of x_1, x_2 and x_3 giving your answers to 4 decimal places. **(3 marks)**

d By considering the change of sign of $f(x)$ in a suitable interval, verify that $\alpha = 0.831$ correct to 3 decimal places. **(2 marks)**

9 $g(x) = e^{x-1} + 2x - 15$

a Show that the equation $g(x) = 0$ can be written as $x = \ln(15 - 2x) + 1$, $x < \frac{15}{2}$ **(2 marks)**

The root of $g(x) = 0$ is α.

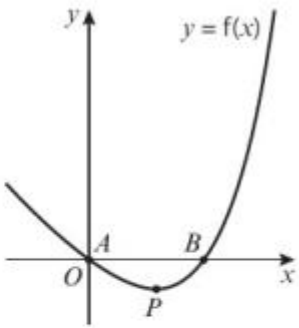
The iterative formula $x_{n+1} = \ln(15 - 2x_n) + 1$, $x_0 = 3$, is used to find a value for α.

b Calculate the values of x_1, x_2 and x_3 to 4 decimal places. **(3 marks)**

c By choosing a suitable interval, show that $\alpha = 3.16$ correct to 2 decimal places. **(3 marks)**

Homework Exercise

10 The diagram shows a sketch of part of the curve with equation $y = f(x)$, where $f(x) = xe^x - 4x$. The curve cuts the x-axis at the points A and B and has a minimum turning point at P, as shown in the diagram.



a Work out the coordinates of A and the coordinates of B. **(3 marks)**

b Find $f'(x)$. **(3 marks)**

c Show that the x-coordinate of P lies between 0.7 and 0.8. **(2 marks)**

d Show that the x-coordinate of P is the solution to the equation $x = \ln\left(\frac{4}{x+1}\right)$. **(3 marks)**

To find an approximation for the x-coordinate of P, the iterative formula $x_{n+1} = \ln\left(\frac{4}{x_n + 1}\right)$ is used.

e Let $x_0 = 0$. Find the values of x_1, x_2, x_3 and x_4. Give your answers to 3 decimal places. **(3 marks)**

Homework Answers

1 **a** **i** $x^2 - 6x + 2 = 0 \Rightarrow 6x = x^2 + 2 \Rightarrow x = \dfrac{x^2 + 2}{6}$

ii $x^2 - 6x + 2 = 0 \Rightarrow x^2 = 6x - 2 \Rightarrow x = \sqrt{6x - 2}$

iii $x^2 - 6x + 2 = 0 \Rightarrow x - 6 + \dfrac{2}{x} = 0 \Rightarrow x = 6 - \dfrac{2}{x}$

b **i** $x = 0.354$ **ii** $x = 5.646$ **iii** $x = 5.646$

c $a = 3, b = 7$

2 **a** **i** $x^2 - 5x - 3 = 0 \Rightarrow x^2 = 5x + 3 \Rightarrow x = \sqrt{5x + 3}$

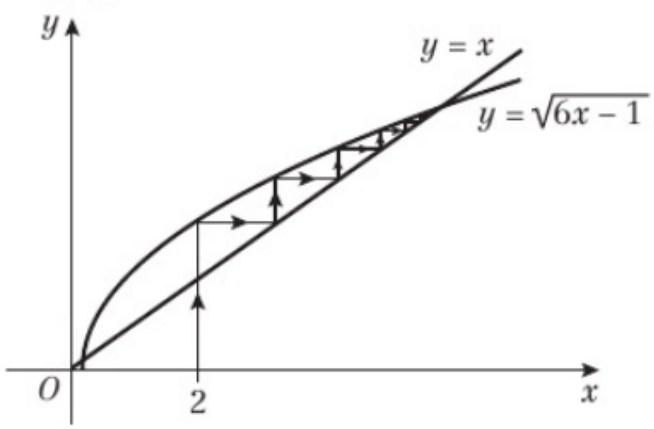
ii $x^2 - 5x - 3 = 0 \Rightarrow x^2 - 3 = 5x \Rightarrow x = \dfrac{x^2 - 3}{5}$

b **i** 5.5 (1 d.p.) **ii** −0.5 (1 d.p.)

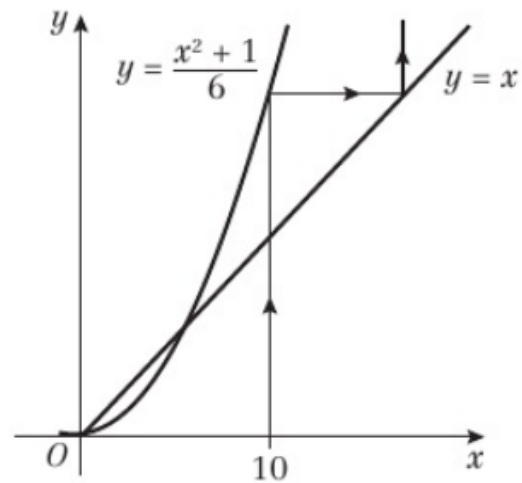
3 **a** $x^2 - 6x + 1 = 0 \Rightarrow x^2 = 6x - 1 \Rightarrow x = \sqrt{6x - 1}$

c The graph shows there are two roots of $f(x) = 0$

b, d



3 e



4 **a** $xe^{-x} - x + 2 = 0 \Rightarrow e^{-x} = \dfrac{x - 2}{x} \Rightarrow e^{x} = \dfrac{x}{x - 2}$

$\Rightarrow x = \ln\left|\dfrac{x}{x - 2}\right|$

b $x_1 = -1.10,\ x_2 = -1.04,\ x_3 = -1.07$

5 **a** **i** $x^3 + 5x^2 - 2 = 0 \Rightarrow x^3 = 2 - 5x^2 \Rightarrow x = \sqrt[3]{2 - 5x^2}$

ii $x^3 + 5x^2 - 2 = 0 \Rightarrow x + 5 - \dfrac{2}{x^2} = 0 \Rightarrow x = \dfrac{2}{x^2} - 5$

iii $x^3 + 5x^2 - 2 = 0 \Rightarrow 5x^2 = 2 - x^3 \Rightarrow x^2 = \dfrac{2 - x^3}{5}$

$\Rightarrow x = \sqrt{\dfrac{2 - x^3}{5}}$

b $x = -4.917$ **c** $x = 0.598$

d It is not possible to take the square root of a negative number over $\mathbb{R}$.

Homework Answers

6 **a** $x^4 - 3x^3 - 6 = 0 \Rightarrow \frac{1}{3}x^4 - x^3 - 2 = 0$

$\Rightarrow \frac{1}{3}x^4 - 2 = x^3 \Rightarrow x = \sqrt[3]{\frac{1}{3}x^4 - 2} \Rightarrow p = \frac{1}{3}, q = -2$

b $x_1 = -1.260$, $x_2 = -1.051$, $x_3 = -1.168$

c $f(-1.1315) = -0.014\ldots < 0$, $f(-1.1325) = 0.0024\ldots > 0$
There is a sign change in this interval, which implies $\alpha = -1.132$ correct to 3 decimal places.

7 **a** $3\cos(x^2) + x - 2 = 0 \Rightarrow \cos(x^2) = \frac{2-x}{3}$

$\Rightarrow x^2 = \arccos\left(\frac{2-x}{3}\right) \Rightarrow x = \left[\arccos\left(\frac{2-x}{3}\right)\right]^{\frac{1}{2}}$

b $x_1 = 1.109$, $x_2 = 1.127$, $x_3 = 1.129$

c $f(1.12975) = 0.000423\ldots > 0$,
$f(1.12985) = -0.0001256\ldots < 0$. There is a sign change in this interval, which implies $\alpha = 1.1298$ correct to 4 decimal places.

8 **a** $f(0.8) = 0.484\ldots$, $f(0.9) = -1.025\ldots$. There is a change of sign in the interval, so there must exist a root in the interval, since f is continuous over the interval.

b $\frac{4\cos x}{\sin x} - 8x + 3 = 0 \Rightarrow 8x = \frac{4\cos x}{\sin x} + 3$

$\Rightarrow x = \frac{\cos x}{2\sin x} + \frac{3}{8}$

c $x_1 = 0.8142$, $x_2 = 0.8470$, $x_3 = 0.8169$

d $f(0.8305) = 0.0105\ldots > 0$, $f(0.8315) = -0.0047\ldots < 0$.
There is a change of sign in the interval, so there must exist a root in the interval.

9 **a** $e^{x-1} + 2x - 15 = 0 \Rightarrow e^{x-1} = 15 - 2x$
$\Rightarrow x - 1 = \ln(15 - 2x)$
$\Rightarrow x = \ln(15 - 2x) + 1$

b $x_1 = 3.1972$, $x_2 = 3.1524$, $x_3 = 3.1628$

c $f(3.155) = -0.062\ldots < 0$, $f(3.165) = 0.044\ldots > 0$.
There is a sign change in this interval, which implies $\alpha = 3.16$ correct to 2 decimal places.

10 **a** $A(0, 0)$ and $B(\ln 4, 0)$

b $f'(x) = xe^x + e^x - 4 = e^x(x + 1) - 4$

c $f'(0.7) = -0.5766\ldots < 0$, $f'(0.8) = 0.0059\ldots > 0$.
There is a sign change in this interval, which implies $f'(x) = 0$ in this range. $f'(x) = 0$ at a turning point.

d $e^x(x + 1) - 4 = 0 \Rightarrow e^x = \frac{4}{x+1} \Rightarrow x = \ln\left(\frac{4}{x+1}\right)$

e $x_1 = 1.386$, $x_2 = 0.517$, $x_3 = 0.970$, $x_4 = 0.708$